EXPERIMENT:2 UPLOAD AND ANALYZE THE DATA SET GIVEN IN CSV FORMAT AND PERFORM DATA PREPROCESSING AND VISUALIZATION.

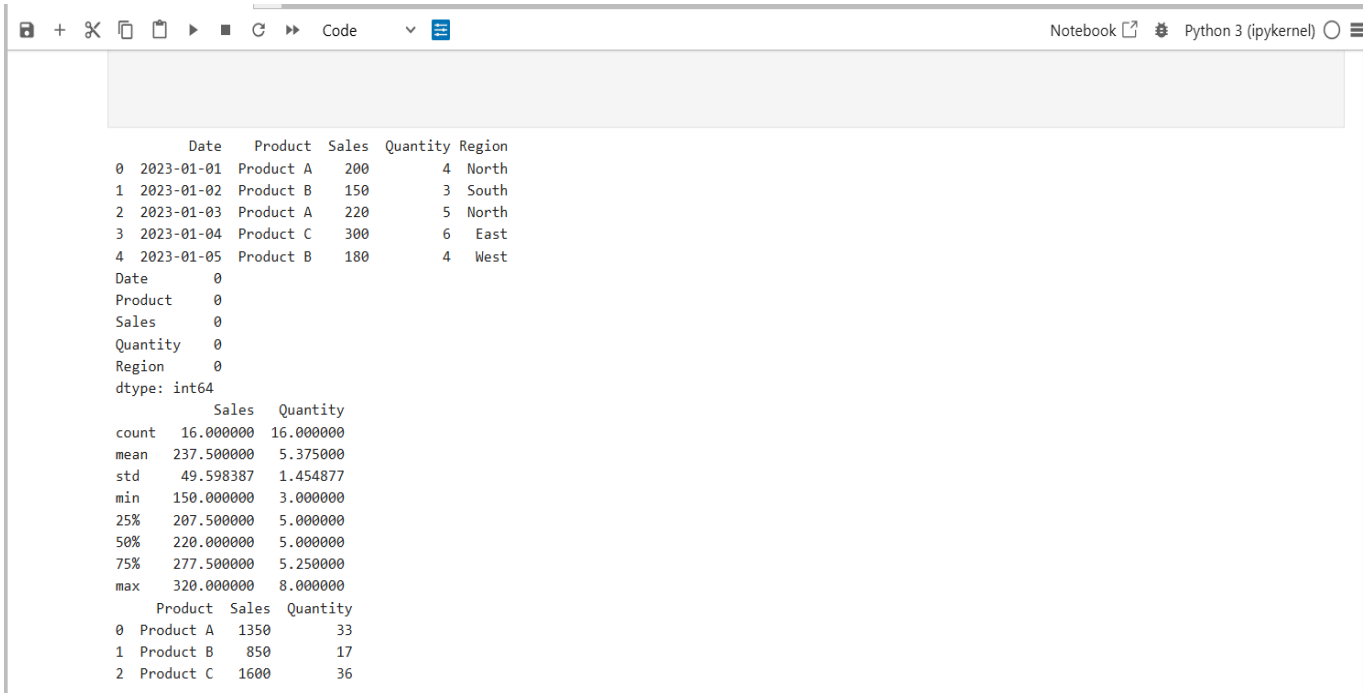
CODE:

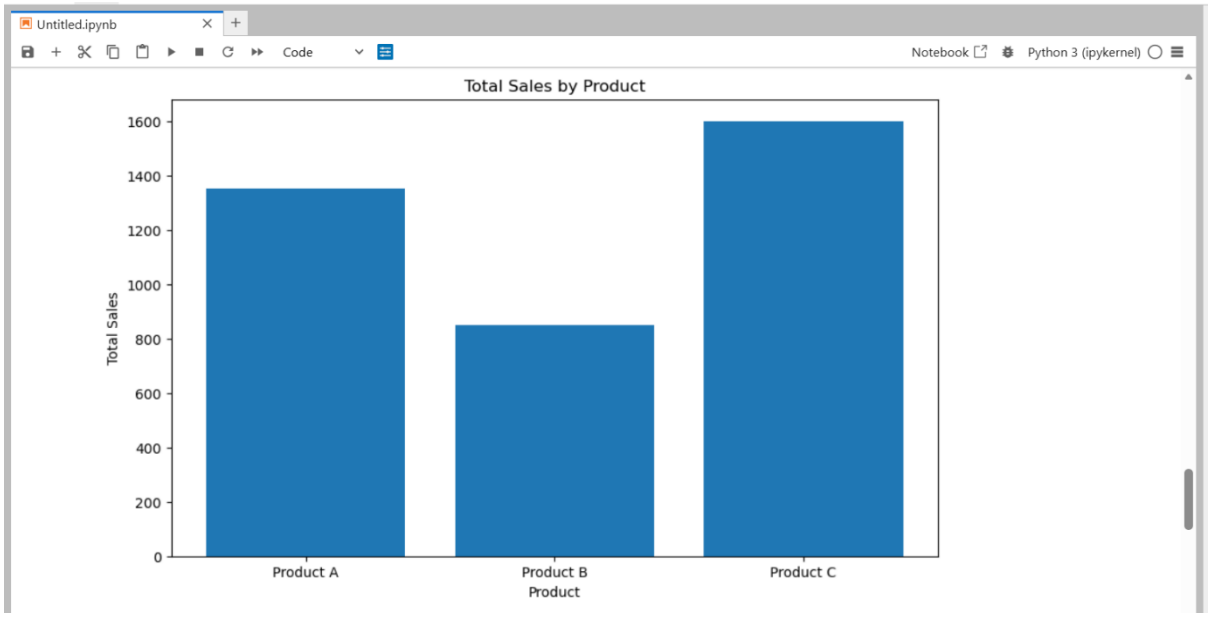
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
# Load the data into a pandas DataFrame
file_path='C:\sales_data.csv'
df = pd.read_csv(file_path)
# Display the first few rows of the DataFrame
print(df.head())
# Check for missing values
print(df.isnull().sum())
# Fill or drop missing values if necessary
df['Sales'].fillna(df['Sales'].mean(), inplace=True)
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)
# Summary statistics
print(df.describe())
# Group by product and calculate the total sales and quantity
product_summary = df.groupby('Product').agg({
'Sales': 'sum',
'Quantity': 'sum'
```

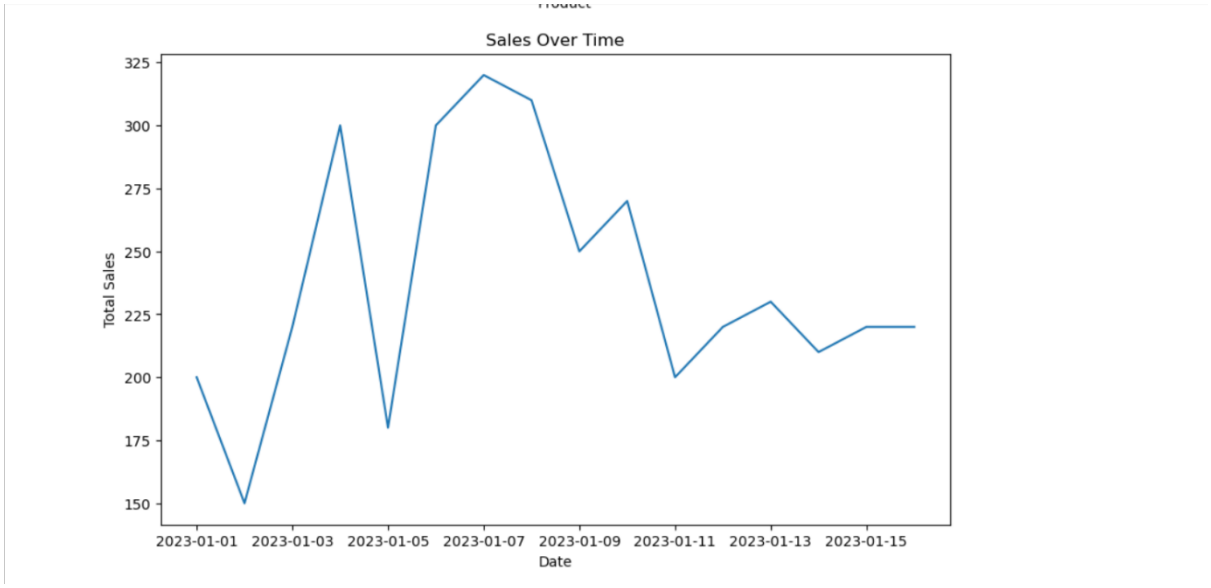
```
}).reset_index()
print(product_summary)
# Bar plot of total sales by product
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()
# Line plot of sales over time
df['Date'] = pd.to_datetime(df['Date'])
sales_over_time = df.groupby('Date').agg({'Sales':
'sum'}).reset_index()
plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'],sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('SalesOver Time')
plt.show()
# Pivot table to analyze sales by region and product
pivot_table = df.pivot_table(values='Sales', index='Region',
columns='Product',
aggfunc=np.sum, fill_value=0)
```

```
print(pivot_table)
# Correlation matrix
correlation_matrix = df.corr()
print(correlation_matrix)
# Heatmap of the correlation matrix
import seaborn as sns
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```

OUTPUT :

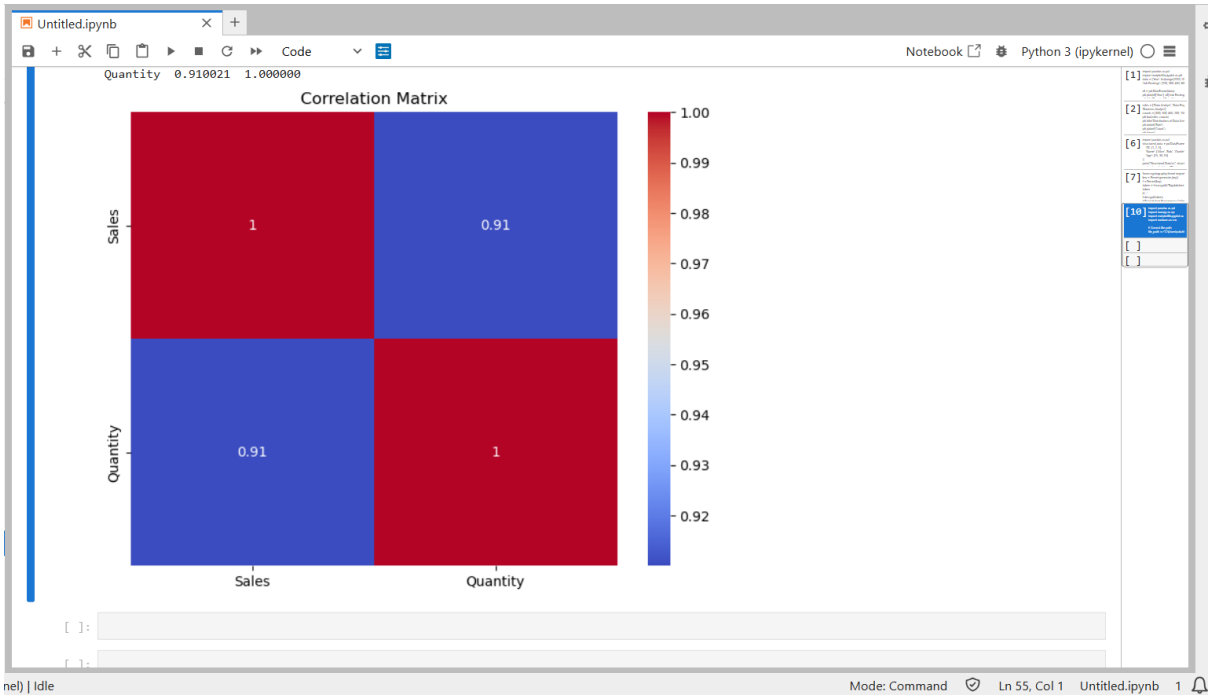






	sum			mean		
Product	Product A	Product B	Product C	Product A	Product B	Product C
Region						
East	310	0	740	310.0	0.0	246.666667
North	720	0	410	240.0	0.0	205.000000
South	320	420	220	320.0	210.0	220.000000
West	0	430	230	0.0	215.0	230.000000

	Sales	Quantity
Sales	1.000000	0.910021
Quantity	0.910021	1.000000


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Code
Notebook
Python 3 (ipykernel)
Quantity 0.910021 1.000000
Correlation Matrix
Sales
Quantity
1
0.91
0.91
1
1.00
0.99
0.98
0.97
0.96
0.95
0.94
0.93
0.92
Mode: Command
Ln 55, Col 1
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